

Prediction Markets as DeGroot Dynamics with Time-Varying Communication Topology

Model Formulation

ECE 5260 / ORIE 5735 — Graph Data Science

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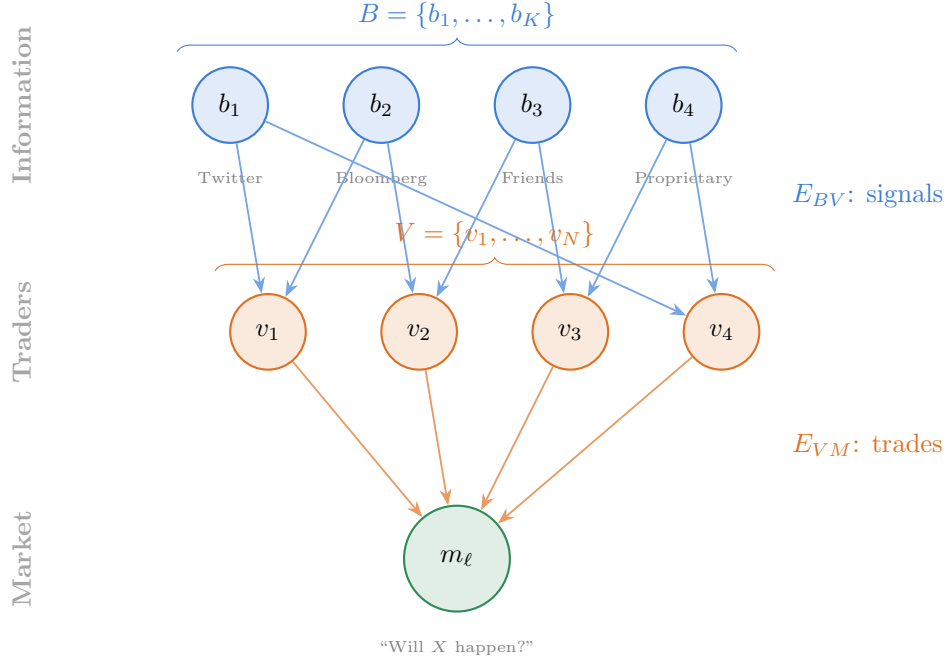
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1 Starting Point: The Tripartite Market Graph

We begin with a prediction market for a binary event with true outcome $\theta \in \{0, 1\}$. The market environment is represented as a directed graph with three layers of nodes and edges connecting them.

1.1 The Three Layers

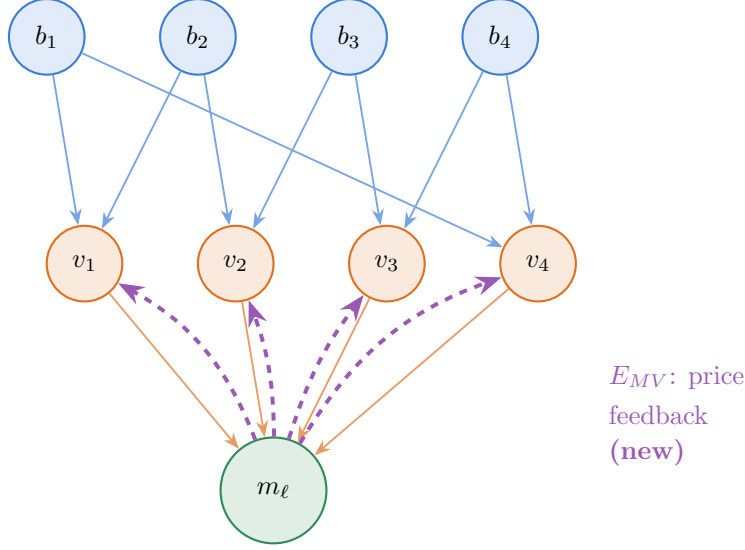


- **Information sources** $B = \{b_1, \dots, b_K\}$: external signals about the event outcome. These are Twitter feeds, news outlets, Bloomberg terminals, conversations with friends, proprietary data — anything that shapes a trader’s beliefs. Each source emits a noisy signal correlated with the true state θ .
- **Traders** $V = \{v_1, \dots, v_N\}$: market participants. Each trader selectively attends to a subset of information sources (their “information diet”) and places bets in the market when they believe the price is wrong.
- **Market** m_ℓ : a prediction market for a binary event $\theta_\ell \in \{0, 1\}$, with a market price $\pi_\ell(t) \in [0, 1]$ that represents the current crowd-implied probability.

So far, information flows in one direction: sources emit signals, traders read them and form beliefs, traders place bets that move the market price. But this picture is incomplete.

2 The Missing Arrow: Price Feedback

In a real market, traders do not just push information into prices — they also *read* prices and update their beliefs accordingly. When a trader sees the market price at 0.72, she treats it as a signal: other people who presumably know things she does not have collectively pushed the price to that level. This creates a feedback loop.



The dashed purple arrows represent the **price feedback channel**: each trader observes the current market price $\pi_\ell(t)$ and incorporates it into their belief update. The graph is now $B \rightarrow V \rightleftharpoons M$, with bidirectional edges between traders and the market.

Key Insight

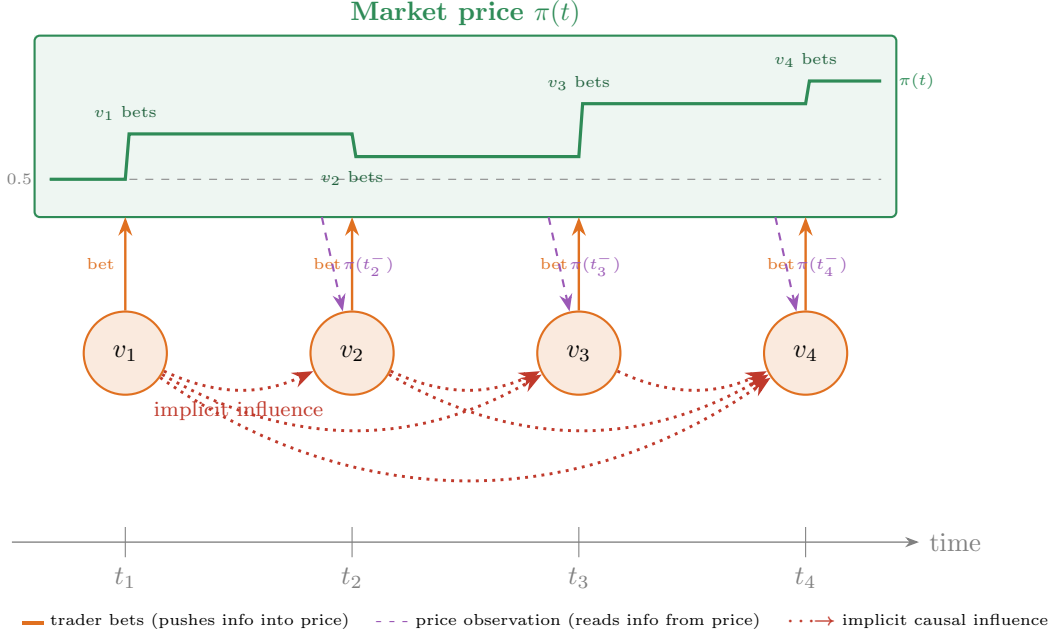
The price feedback channel changes the model fundamentally. Without it ($E_{MV} = \emptyset$), the market is a one-shot aggregation mechanism: traders form beliefs from sources and trade once. With it, the market becomes a *dynamical system* where beliefs and prices co-evolve over time.

3 The Hidden Consequence: Implicit Trader-to-Trader Edges

The price feedback channel has a subtle but crucial consequence. Consider what happens when trader v_2 observes the market price at time t :

1. Trader v_1 bet at time $t_1 < t$, pushing the price from $\pi(t_1^-)$ to $\pi(t_1^+)$.
2. The price $\pi(t)$ that trader v_2 sees *encodes* trader v_1 's belief (filtered through the market mechanism).
3. When v_2 incorporates $\pi(t)$ into her belief, she is *indirectly influenced* by v_1 .

This means the price acts as a **communication channel** between traders, creating *implicit* edges in the trader layer:



Key Insight

Even though there are no explicit $V \rightarrow V$ edges in the tripartite graph, the price feedback channel creates **implicit causal influence** between traders. The influence is *directed* (only past bets affect future traders) and *time-varying* (the set of influential traders grows as more people bet). This is the observation that motivates the DeGroot reformulation.

Moreover, the strength of this implicit influence is *not equal* across traders. A trader who bet a large quantity (low risk aversion a_i) moved the price more, and therefore has more implicit influence on subsequent traders. This is captured precisely by the price impact rule: trader i 's bet changes the price by $\gamma_\ell \cdot q_i = \gamma_\ell(p_i - \pi)/a_i$, so the influence of trader i on the price is proportional to γ_ℓ/a_i .

4 Formal Setup: State Space and Parameters

Before writing the dynamics, we define all quantities precisely.

4.1 Information Sources (Stubborn Agents)

There are K information sources. Each source b_j emits a signal at each time step:

$$s_j(t) = \theta + \mu_j + \varepsilon_j(t), \quad (1)$$

where $\theta \in \{0, 1\}$ is the true event outcome, $\mu_j \in \mathbb{R}$ is a *persistent bias* (e.g., partisan lean), and $\varepsilon_j(t)$ is zero-mean noise with cross-source covariance $\text{Cov}(\varepsilon_j, \varepsilon_k) = [\Sigma_B]_{jk}$.

Sources do not observe the market or each other. In the DeGroot framework, they are **stubborn agents**: their signals are determined exogenously and do not update in response to other nodes in the graph.

4.2 Traders (Regular Agents)

There are N traders. Each trader v_i is parameterized by:

Symbol	Name	Meaning
$\mathbf{w}_i \in \mathbb{R}_{\geq 0}^K$	Attention vector	How much trader i attends to each source. Entry w_{ij} is the weight on source b_j . This defines the $B \rightarrow V$ edges.
$a_i > 0$	Risk aversion	Scales position size inversely: more risk-averse traders bet smaller.
$\lambda_i > 0$	Arrival rate	Rate of a Poisson process governing when trader i checks the market.
$\alpha_i \in (0, 0.5]$	Deadband	Trader i only bets when $ p_i(t) - \pi(t) > \alpha_i$ (minimum perceived edge).
$\beta_i^{\text{info}} \in [0, 1]$	Information weight	How much trader i trusts their own information sources.
$\beta_i^{\text{price}} \in [0, 1]$	Price weight	How much trader i trusts the market price as a signal.

The constraint $\beta_i^{\text{info}} + \beta_i^{\text{price}} \leq 1$ ensures the residual weight $(1 - \beta_i^{\text{info}} - \beta_i^{\text{price}})$ on the prior belief is non-negative.

4.3 Market

The market m_ℓ has a price $\pi_\ell(t) \in [0, 1]$ initialized to $\pi_\ell(0) = 0.5$ (uninformative prior) and a depth parameter $\gamma_\ell > 0$ controlling price impact.

4.4 Augmented State Vector

Following the DeGroot convention, we stack sources and traders into a single state vector:

$$\mathbf{x}(t) = \begin{pmatrix} \mathbf{s}(t) \\ \mathbf{p}(t) \end{pmatrix} \in \mathbb{R}^{K+N}, \quad (2)$$

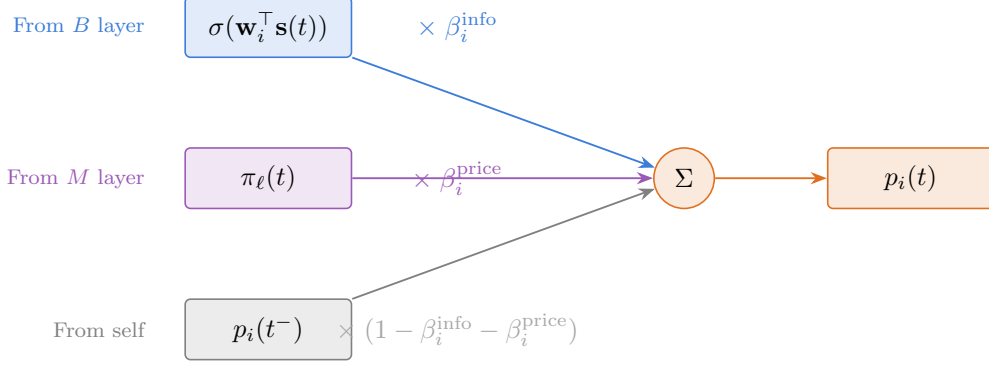
where $\mathbf{s}(t) = (s_1(t), \dots, s_K(t))^\top$ collects the source signals and $\mathbf{p}(t) = (p_1(t), \dots, p_N(t))^\top$ collects the trader beliefs. This is the state that the DeGroot dynamics will operate on.

5 Step 1: Trader Belief Update Rule

When trader i arrives at time t (drawn from their Poisson process with rate λ_i), they form an updated belief about the event by combining three components:

$$p_i(t) = \underbrace{\beta_i^{\text{info}} \cdot \sigma(\mathbf{w}_i^\top \mathbf{s}(t))}_{\text{information from sources}} + \underbrace{\beta_i^{\text{price}} \cdot \pi_\ell(t)}_{\text{information from price}} + \underbrace{(1 - \beta_i^{\text{info}} - \beta_i^{\text{price}}) \cdot p_i(t^-)}_{\text{prior belief (inertia)}} \quad (3)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic sigmoid (mapping the weighted signal sum into $[0, 1]$), and t^- denotes the time of trader i 's previous arrival.



This is a convex combination: the three weights sum to 1, so the updated belief $p_i(t)$ is a weighted average of what the trader's sources say, what the market says, and what the trader previously believed.

6 Step 2: From Belief to Trade

After updating her belief, trader i decides whether to bet and how much.

6.1 Demand Function

The trader's demand (signed bet quantity) is:

$$q_i(t) = \begin{cases} \frac{p_i(t) - \pi_\ell(t)}{a_i}, & \text{if } |p_i(t) - \pi_\ell(t)| > \alpha_i, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The deadband α_i ensures traders only bet when their perceived edge exceeds a threshold. This creates a **selection effect**: aggressive traders (small α_i) bet first, shaping the early price trajectory for everyone who comes after.

6.2 Price Impact

When trader i bets, the market maker absorbs the order and adjusts the price:

$$\pi_\ell(t^+) = \text{clip}_{[0,1]}(\pi_\ell(t) + \gamma_\ell \cdot q_i(t)). \quad (5)$$

The depth parameter γ_ℓ controls how much a single trade moves the price. Small γ_ℓ means a deep, liquid market; large γ_ℓ means a thin market where individual bets have large impact.

7 Step 3: Unrolling the Price — Where the DeGroot Structure Emerges

This is the key step. By unrolling the price impact equation, we can see exactly how past traders' beliefs are embedded in the current price.

7.1 The Price as a Weighted Average of Past Beliefs

Let the sequence of traders who have bet up to time t be $i(1), i(2), \dots, i(n)$ at times $t_1 < t_2 < \dots < t_n < t$. Ignoring the clipping for clarity, the price at time t is:

$$\pi_\ell(t) = \pi_\ell(0) + \gamma_\ell \sum_{\tau=1}^n \frac{p_{i(\tau)}(t_\tau) - \pi_\ell(t_\tau)}{a_{i(\tau)}}. \quad (6)$$

This can be reorganized (via a telescoping argument) to show that the price is approximately a *risk-aversion-weighted running average* of past beliefs:

$$\pi_\ell(t) \approx \frac{\sum_{\tau=1}^n \frac{1}{a_{i(\tau)}} p_{i(\tau)}(t_\tau)}{\sum_{\tau=1}^n \frac{1}{a_{i(\tau)}}}. \quad (7)$$

Key Insight

The market price at time t is a precision-weighted average of the beliefs of all traders who have bet so far, where the “precision” of each trader is $1/a_i$ (the inverse of their risk aversion). When trader v_j observes this price and weights it by β_j^{price} , she is implicitly placing weight on every previous trader’s belief, proportional to how much that trader moved the price.

7.2 Implicit Peer Influence

Now substitute the price expression into the belief update. When a new trader v_j arrives at time t and observes $\pi_\ell(t)$:

$$\begin{aligned} p_j(t) &= \beta_j^{\text{info}} \cdot \sigma(\mathbf{w}_j^\top \mathbf{s}(t)) + \beta_j^{\text{price}} \cdot \pi_\ell(t) + (1 - \beta_j^{\text{info}} - \beta_j^{\text{price}}) \cdot p_j(t^-) \\ &\approx \beta_j^{\text{info}} \cdot \sigma(\mathbf{w}_j^\top \mathbf{s}(t)) + \beta_j^{\text{price}} \cdot \underbrace{\sum_{k \in \mathcal{A}(t)} \frac{\gamma_\ell/a_k}{\sum_{k' \in \mathcal{A}(t)} \gamma_\ell/a_{k'}} \cdot p_k(t_k)}_{\text{weighted sum of past traders' beliefs}} + (1 - \beta_j^{\text{info}} - \beta_j^{\text{price}}) \cdot p_j(t^-), \end{aligned} \quad (8)$$

where $\mathcal{A}(t) = \{i(\tau) : \tau < t, \text{ trader } i(\tau) \text{ bet}\}$ is the set of active (already-bet) traders.

This is a weighted average of other traders’ beliefs — exactly the form of a DeGroot update.

8 Step 4: The DeGroot Block Form

We can now write the full dynamics in matrix form. Define:

Definition 1 (Effective Peer-Influence Matrix). $D(t) \in \mathbb{R}^{N \times N}$ with entries:

$$[D(t)]_{jk} = \begin{cases} 1 - \beta_j^{\text{info}} - \beta_j^{\text{price}}, & \text{if } k = j \text{ (self-weight),} \\ \beta_j^{\text{price}} \cdot \frac{\gamma_\ell/a_k}{\sum_{k' \in \mathcal{A}(t)} \gamma_\ell/a_{k'}}, & \text{if } k \in \mathcal{A}(t), k \neq j, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Definition 2 (Effective Source-Influence Matrix). $B(t) \in \mathbb{R}^{N \times K}$ with entries:

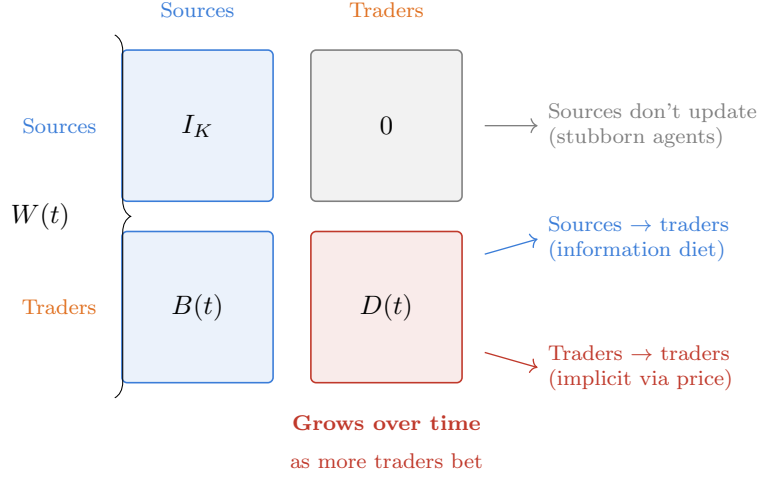
$$[B(t)]_{jk} = \beta_j^{\text{info}} \cdot w_{jk} \cdot \sigma'(\mathbf{w}_j^\top \mathbf{s}(t)), \quad (10)$$

where $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ is the logistic derivative. In the linearized regime (beliefs near 0.5), $\sigma' \approx 0.25$, so B simplifies to $0.25 \cdot \beta_j^{\text{info}} \cdot w_{jk}$.

Proposition 1 (DeGroot Block Form). *In the linearized regime, the augmented dynamics are:*

$$\underbrace{\begin{pmatrix} \mathbf{s}(t+1) \\ \mathbf{p}(t+1) \end{pmatrix}}_{\mathbf{x}(t+1)} = \underbrace{\begin{pmatrix} I_K & 0_{K \times N} \\ B(t) & D(t) \end{pmatrix}}_{W(t)} \underbrace{\begin{pmatrix} \mathbf{s}(t) \\ \mathbf{p}(t) \end{pmatrix}}_{\mathbf{x}(t)} + \begin{pmatrix} \Delta \boldsymbol{\varepsilon}(t) \\ \mathbf{0}_N \end{pmatrix}. \quad (11)$$

This is the classical **DeGroot model with stubborn agents** (Friedkin–Johnsen / Ghaderi–Srikant), with one crucial difference: $W(t)$ is *time-varying*.



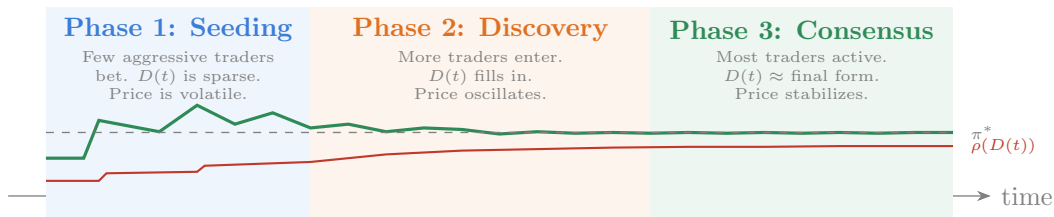
Important Modeling Choice

The peer-influence block $D(t)$ is *not fixed*. It starts nearly empty (only self-weights on the diagonal) when no one has bet, and fills in as traders cross the participation threshold. The communication topology grows over the life of the market. This is what makes the model a *dynamic* DeGroot system rather than a classical static one.

9 Step 5: The Time-Varying Topology

The evolution of $D(t)$ over the life of a market follows a characteristic pattern.

9.1 Three Phases of Market Life



1. **Phase 1 — Seeding** (early market). Only a few aggressive traders (small deadband α_i) have bet. The matrix $D(t)$ is mostly diagonal (traders are influenced mainly by their own prior and sources). The price is volatile because each individual bet has large impact on a thin market. The spectral radius $\rho(D(t))$ is low.

2. **Phase 2 — Discovery** (active trading). As the price moves away from 0.5, more traders' beliefs cross the deadband threshold, triggering bets. $D(t)$ fills in rapidly. The price oscillates as traders with different information update and trade. $\rho(D(t))$ increases, reflecting growing peer influence through the price channel.
3. **Phase 3 — Consensus** (late market). Most traders have bet. $D(t)$ has approximately reached its final form. The price stabilizes near the equilibrium value. New arrivals find the price close to their belief and do not bet (deadband not crossed), so the topology stops growing.

9.2 The Selection Effect

Remark 1 (Who Bets First Shapes the Market). *Because aggressive traders (small α_i) bet first, the early price trajectory $\pi_\ell(t)$ for small t is shaped by the high-conviction subpopulation. Subsequent traders who trust the price signal (β_i^{price} large) effectively inherit the information — and biases — of this early cohort. This creates a **temporal attention asymmetry**: early bets have outsized influence on the eventual equilibrium, regardless of whether those early bettors had good information.*

This is a testable prediction of the model: markets where early traders are well-informed should converge faster and more accurately than markets where early traders are poorly informed or biased, even if the total information in the trader population is identical.

10 Convergence and Steady State

10.1 Fixed-Topology Benchmark

If the topology were frozen (all traders active from $t = 0$, so $D(t) \equiv D$ is constant), the DeGroot model with stubborn agents converges to:

$$\mathbf{p}^* = (I_N - D)^{-1} B \mathbf{s}^* \quad (12)$$

where $\mathbf{s}^* = \mathbb{E}[\mathbf{s}(t)] = \boldsymbol{\theta} + \boldsymbol{\mu}$ is the long-run source signal vector. The matrix $(I_N - D)^{-1}$ exists whenever $\rho(D) < 1$, which holds as long as every trader retains some weight on either information or self (i.e., $\beta_i^{\text{price}} < 1$).

The terminal beliefs $\mathbf{p}^*$ are determined entirely by:

- The source-influence matrix B (who reads what),
- The peer-influence matrix D (implicit price-mediated influence), and
- The long-run source signals $\mathbf{s}^*$ (truth plus bias).

10.2 Convergence Speed

The rate of convergence is controlled by the spectral radius of D :

$$\tau_{\text{mix}} \sim \frac{1}{1 - \rho(D)} \sim \frac{1}{1 - \bar{\beta}^{\text{price}}}, \quad (13)$$

where $\bar{\beta}^{\text{price}}$ is the population-average price trust. This yields two limiting cases:

- **Low price trust** ($\bar{\beta}^{\text{price}} \rightarrow 0$): fast convergence. Each trader relies on private information, so the market quickly aggregates independent signals. But the price doesn't help spread information across traders.

- **High price trust** ($\bar{\beta}^{\text{price}} \rightarrow 1$): slow convergence. Traders are mostly echoing the market price back to each other, creating an echo-chamber dynamic. New private information enters slowly.

This suggests an **optimal interior value** of $\bar{\beta}^{\text{price}}$ that balances the benefit of social learning (traders learn from each other via the price) against the cost of echo-chamber effects (traders stop bringing independent information).

10.3 Time-Varying Topology

With the growing topology, convergence is governed by the *joint spectral radius* of the sequence $\{D(t)\}_{t=0}^T$. A sufficient condition for convergence is that $\rho(D(t)) < 1$ for all t , which holds whenever $\beta_i^{\text{price}} < 1$ for all i .

11 The Log-Odds Coordinate System

The belief update in Eq. (3) involves a sigmoid nonlinearity $\sigma(\mathbf{w}_i^\top \mathbf{s}(t))$, which makes the source-influence matrix $B(t)$ state-dependent (through σ'). To obtain an *exact* linear system, we change coordinates to log-odds space.

Define:

$$\ell_i(t) = \log \frac{p_i(t)}{1 - p_i(t)}, \quad \Lambda_\ell(t) = \log \frac{\pi_\ell(t)}{1 - \pi_\ell(t)}. \quad (14)$$

In this coordinate system, the belief update becomes:

$$\ell_i(t) = \beta_i^{\text{info}} \cdot \mathbf{w}_i^\top \mathbf{s}(t) + \beta_i^{\text{price}} \cdot \Lambda_\ell(t) + (1 - \beta_i^{\text{info}} - \beta_i^{\text{price}}) \cdot \ell_i(t^-). \quad (15)$$

This is exactly linear — no sigmoid, no state-dependent derivative. The DeGroot block form $\ell(t+1) = W(t)\ell(t)$ holds *exactly*, not as an approximation.

Key Insight

Working in log-odds space eliminates the sigmoid nonlinearity and makes the DeGroot structure exact. This is the natural coordinate system for the model: beliefs near 0 or 1 produce large log-odds, reflecting strong conviction, while beliefs near 0.5 produce log-odds near zero, reflecting uncertainty.

12 The Information Matrix Φ

Independent of the dynamic price-mediated influence, traders' beliefs are correlated through their shared consumption of information sources. This static correlation structure is captured by the **information matrix**.

The attention weights assemble into a biadjacency matrix $W \in \mathbb{R}_{\geq 0}^{K \times N}$ with $W_{ji} = w_{ij}$. The information matrix is:

$$\Phi = W^\top \Sigma_B^{-1} W \in \mathbb{R}^{N \times N} \quad (16)$$

12.1 What Φ Captures

To build intuition, consider the entry Φ_{ik} :

$$\Phi_{ik} = \sum_{j=1}^K \sum_{j'=1}^K w_{ij} [\Sigma_B^{-1}]_{jj'} w_{kj'}. \quad (17)$$

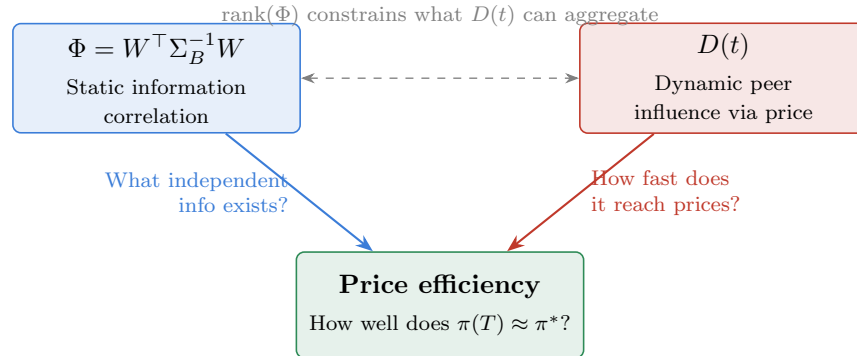
This is the *precision-weighted overlap* between trader i 's and trader k 's information diets. It accounts for the correlation structure of the sources: two traders who read highly correlated sources (e.g., news outlets that copy from the same wire service) get less credit for “shared information” than two traders who read independent sources.

12.2 Spectral Interpretation

Spectral property	Economic interpretation
Φ_{ii} (diagonal)	Total precision of trader i 's aggregate signal. Standard result from estimation theory.
$\text{rank}(\Phi)$	Number of independent “belief directions” in the trader population. If all traders read the same sources with similar weights, $\text{rank}(\Phi) \approx 1$ regardless of N .
Eigenvectors of Φ	Clusters of traders who will form correlated beliefs — the groups that will emerge as opposing camps, even without direct communication.
Eigenvalue gaps	How sharply separated the belief clusters are. A large gap between λ_1 and λ_2 means one dominant information axis; small gaps mean diffuse, multi-polar information.

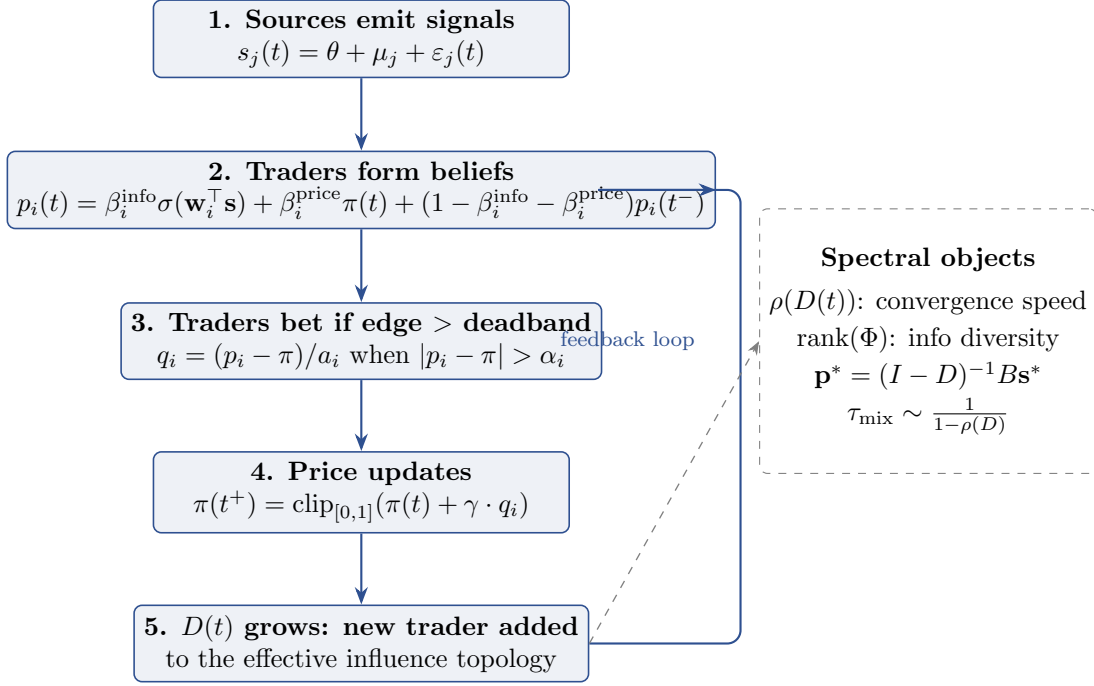
12.3 Relationship to $D(t)$

The information matrix Φ characterizes the *static* correlation structure induced by shared information consumption (the $B \rightarrow V$ layer). The peer-influence matrix $D(t)$ characterizes the *dynamic* influence structure induced by the price feedback channel (the $V \rightleftharpoons M$ loop). Together, they determine the full dynamics of the market:



The central claim of this model: **price efficiency depends on both objects.** A market with high $\text{rank}(\Phi)$ (diverse information) but high $\rho(D)$ (echo-chamber price dynamics) aggregates slowly. A market with low $\text{rank}(\Phi)$ (everyone reading the same thing) converges fast but to an estimate that reflects limited independent information. The best outcomes require both diverse information *and* effective dynamic aggregation.

13 Summary: The Complete Picture



Steps 2–5 repeat at each trader arrival (Poisson process). The DeGroot matrix $W(t)$ encodes the full state of the system at each moment: who is influencing whom, through which channels, and with what strength. Its spectral properties predict whether and how quickly the market converges to an informationally efficient price.